

SOC Investigation Report - Possible Logon Breach

External indicators are defanged for safe handling.

Client	Maple Tax Solutions (MTS)
Prepared by	Graham McKean
Date	August 2026
Classification	CONFIDENTIAL
Severity	High

Findings

Initial Alert Activity	2026-07-26 10:23:43 UTC
Alert Generated	2026-07-26 10:24:41 UTC
Affected Host	mts-dc.mts.local
Internal IP	192.168.10.8
Endpoint External IP	40.70.80[.]252 (host attribute; not attacker/source IP)
Target Account	mts\administrator
Primary Alert-Associated Source IP	176.53.159[.]230
Additional Related Source IP	176.53.159[.]222
Initial Access Service	Remote Desktop Protocol (RDP), TCP/3389
Authentication	Successful NTLM authentication
Interactive Access	RemoteInteractive administrator session confirmed
Detection Source	Microsoft Defender Advanced Threat Protection
MITRE ATT&CK	Credential Access - T1110 (Brute Force), as mapped by the Defender alert
Severity	High
Status at Alert Creation	New
Confirmed Outcome	Successful external RDP access using mts\administrator

SOC Analyst Verdict

Verdict: True Positive - Successful External RDP Logon - High Confidence.

Advanced Hunting confirmed that 176.53.159[.]230 established an accepted inbound TCP connection to RDP port 3389 on mts-dc.mts.local at 10:23:40 UTC. At the same timestamp, mts\administrator successfully authenticated from that IP using NTLM. A second source, 176.53.159[.]222, established another accepted RDP connection at 10:25:15 UTC followed by successful NTLM administrator authentication and a RemoteInteractive session. The evidence confirms successful external RDP access, but telemetry alone does not establish whether the person using the credentials was authorised.

Confidence Matrix

Area	Confidence
Alert detection	High
Host attribution	High
User attribution	High
Source IP attribution	High
Successful authentication	High
RDP service attribution	High
Remote interactive session	High
Post-logon command/tool execution	Low / Not observed
Authorisation of remote access	Not established
Scope confidence	Medium
Overall verdict	High

Observed Indicators

Host	mts-dc.mts.local
Account	mts\administrator
Internal IP	192.168.10.8
Endpoint External IP	40.70.80[.]252
Alert-associated source IP	176.53.159[.]230
Additional related source IP	176.53.159[.]222
Other failed-authentication source IPs	138.226.239[.]7 and 45.227.254[.]153
Service	RDP / TCP 3389
Authentication protocol	NTLM

40.70.80[.]252 is the affected endpoint's recorded external IP and is not treated as the attacker/source IP. The earlier failed-authentication source IPs were observed before the confirmed RDP access, but the evidence does not establish that they were part of the same activity.

Confirmed Incident-Associated Indicators

- 176.53.159[.]230 - accepted inbound RDP connection to mts-dc.mts.local:3389 and successful NTLM authentication as mts\administrator at 10:23:40 UTC.
- 176.53.159[.]222 - accepted inbound RDP connection to mts-dc.mts.local:3389 at 10:25:15 UTC, followed by successful NTLM administrator authentication and a RemoteInteractive session.

These addresses are confirmed incident-associated source indicators. Their external reputation or ownership was not independently established, so they are not labelled confirmed malicious solely on the basis of this telemetry.

Investigation Summary

On 26 July 2026, Microsoft Defender for Endpoint generated a high-severity Possible logon breach alert for mts-dc.mts.local involving mts\administrator. At 10:23:40 UTC, 176.53.159[.]230 established an accepted inbound TCP connection to RDP port 3389 and Defender recorded a successful NTLM network logon for mts\administrator from the same address. The alert activity timestamp followed at 10:23:43 UTC.

At 10:25:15 UTC, 176.53.159[.]222 established another accepted inbound RDP connection. Successful NTLM administrator logons followed at 10:25:21-10:25:24 UTC, and a RemoteInteractive logon was recorded at 10:25:27 UTC. Process telemetry showed normal RDP/interactive-session processes, including rdpclip.exe, and Microsoft Edge activity under the administrator account.

A targeted hunt for common command-line and administrative tools, including cmd.exe, powershell.exe, net.exe, whoami.exe, schtasks.exe, wmic.exe and reg.exe, returned no results in the reviewed period. Successful external RDP access is therefore confirmed, but no obvious command-line or administrative-tool execution was identified. Authorisation remains to be validated with the account/system owner.

Timeline

Time (UTC)	Event
09:56:07	Failed NTLM network logon to mts\administrateur from 138.226.239[.]7 - InvalidUserNameOrPassword.
10:01:02	Failed NTLM network logon to mts\administrator from 45.227.254[.]153 - InvalidUserNameOrPassword.
10:23:40	176.53.159[.]230 accepted inbound TCP connection to mts-dc.mts.local:3389 (RDP).
10:23:40	Successful NTLM network logon for mts\administrator from 176.53.159[.]230.
10:23:43	Defender alert activity timestamp for Possible logon breach.
10:24:41	High-severity Defender alert generated.

10:25:15	176.53.159[.]222 accepted inbound TCP connection to mts-dc.mts.local:3389.
10:25:21-10:25:24	Successful NTLM network logons for mts\administrator from 176.53.159[.]222.
10:25:26	RDP session process rdpclip.exe observed under mts\administrator, initiated by Remote Desktop Services.
10:25:27	RemoteInteractive logon recorded for mts\administrator; unlock events associated with 176.53.159[.]222 also observed.
10:25:34 onward	Interactive-session activity including Microsoft Edge processes observed under mts\administrator.

5Ws and 1H

Who	The affected account was mts\administrator. Confirmed remote source addresses were 176.53.159[.]230 and 176.53.159[.]222. The human operator is not identified.
What	Successful external RDP connections were followed by successful NTLM authentication using the privileged administrator account; a RemoteInteractive session was also recorded.
When	Primary successful connection/authentication at 10:23:40 UTC on 26 Jul 2026; additional successful RDP activity from 10:25:15 UTC.
Where	mts-dc.mts.local, internal IP 192.168.10.8, via Remote Desktop Services on TCP/3389. 40.70.80[.]252 is the endpoint external IP, not the source IP.
Why	Defender correlated infrastructure associated with historical brute-force activity with a successful administrator sign-in. Operator intent and authorisation cannot be established from telemetry alone.
How	External systems connected to TCP/3389 and mts\administrator authenticated successfully using NTLM. The second sequence progressed to a RemoteInteractive logon and normal RDP session processes.

Assessment

The Defender alert is a true positive for successful external RDP access using the privileged mts\administrator account. The evidence confirms the connection, successful NTLM authentication and subsequent interactive session. The targeted post-logon process hunt found no common command-line or administrative-tool execution; this does not prove that no other activity occurred. If MTS confirms the source IPs were not approved infrastructure and the logons were unexpected, the activity should be treated as confirmed unauthorised account compromise.

Recommendations

- Validate with the account/system owner whether the administrator RDP sessions at 10:23-10:25 UTC were authorised and whether the source IPs belong to approved VPN, jump-host, administrative or service-provider infrastructure.
- If the access was unauthorised, contain the affected host, reset the administrator credentials, invalidate active sessions and review other privileged accounts for exposure.
- Preserve the relevant Defender/Advanced Hunting evidence and authentication timeline for incident handling.
- Review the full device timeline and broader telemetry for file, registry, service, scheduled-task, PowerShell, network and persistence activity after the RDP sessions.
- Hunt 176.53.159[.]230 and 176.53.159[.]222 across the wider environment for additional connections, logons or affected devices.
- Restrict public RDP exposure where possible and require controlled remote access through approved VPN/jump-host infrastructure with MFA or equivalent strong authentication.
- Continue centralised collection and retention of Windows Security and Defender telemetry for future authentication investigations.

Investigation Limitations

- The available telemetry does not establish whether the successful RDP sessions were authorised by MTS.
- The human operator behind the external source IPs is not identified.
- The two earlier failed-authentication source IPs cannot be confidently linked to the later successful RDP sessions.
- The targeted post-logon process hunt returned no common command-line or administrative-tool execution, but this does not exclude activity through other processes or mechanisms.
- No independent OSINT reputation assessment of 176.53.159[.]230 or 176.53.159[.]222 is relied upon in this report.
- The investigation scope is limited to the Defender/Advanced Hunting telemetry reviewed for the stated period and host.

Queries Used and Purpose

Hunt	Purpose / Result
DeviceLogonEvents	Validated successful and failed authentication

	around the alert; identified source IPs, account, logon type, protocol and failure reason.
DeviceProcessEvents - administrator activity	Identified processes under the administrator account after successful logon and established evidence of an interactive session.
DeviceNetworkEvents - source IP pivot	Confirmed 176.53.159[.]230 and 176.53.159[.]222 connected inbound to TCP/3389 and were handled by Remote Desktop Services.
Targeted administrative-tool process hunt	Searched for common command-line and administrative tools after the RDP session; no results were returned.